Digestion in duodenum and small intestine: The nucleic acids are hydrolysed to a mixture of nucleotides by ribonucleases and deoxyribonucleases (both endo and exonucleases) present in pancreatic and intestinal secretions. The nucleotides thus formed liberate the phosphate and form nucleosides by the action of nucleotidases. The resulting nucleosides are either absorbed or hydrolyzed by nucleosidases to purine and pyrimidine bases and pentose sugars (Figure 2.23). Some of the unabsorbed purines are metabolized by the intestinal bacteria.

Absorption of nucleic acids: The nucleic acid bases are absorbed by active transport in the small intestine and the pentoses are absorbed with the other sugars.

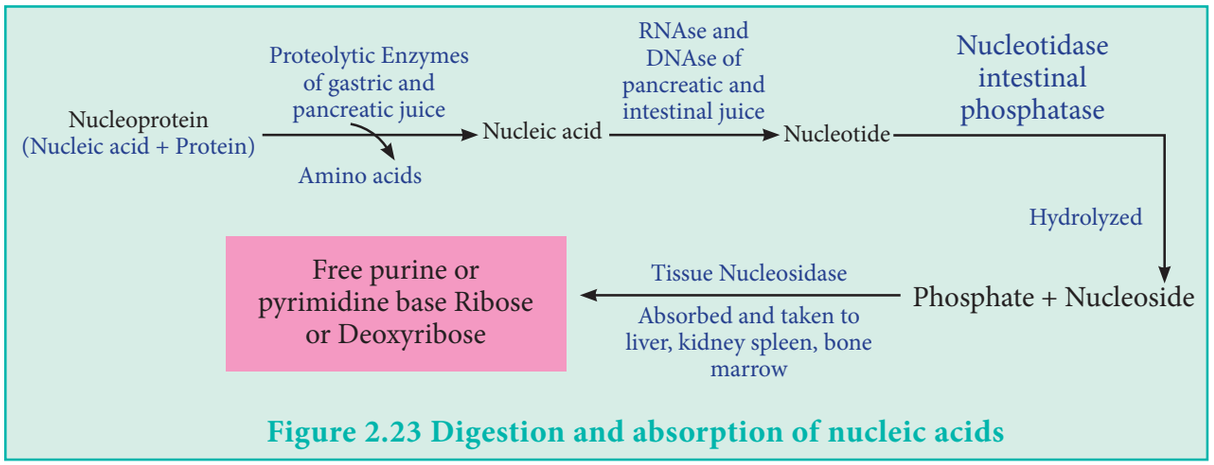

Figure 2.23 Digestion and absorption of nucleic acids

2.3. Gastrointestinal hormones

Table 2.3 Gastrointestinal hormone actions, Stimuli for secretion and site of secretion

Hormone	Stimuli for Secretion	Site of Secretion	Actions
Gastrin	Protein, Distention (Acid inhibits release)	G cells of the antrum, duodenum, and jejunum	Stimulates Gastric acid secretion Mucosal growth
Cholecystokinin	Protein Fat Acid	I cells of the duodenum, jejunum, and ileum	Stimulates Pancreatic enzyme secretion Pancreatic bicarbonate secretion Gallbladder contraction Growth of exocrine pancreas

XII U2 Digestion.indd 50 4/8/2020 3:05:31 PM